gr00t-arm

A desk robot arm that learns from demonstrations with NVIDIA GR00T

How it connects to GR00T, and how to build, wire, calibrate, train and run it

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Arm: *Easy Robot Arm* by mobiobi (Printables 97370), CC BY-NC-SA, used unmodified

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Part I
How the arm connects to GR00T

1 The idea in one picture

GR00T is NVIDIA's *robot foundation model*: one neural network, pretrained on a huge amount of robot and human data, that turns **what a robot sees**, **what it is told** and **where its joints are** into **the next motions**. It was built for humanoids, but it is designed so a *new* body can be taught to it with a small amount of data. This arm is that new body: five $2 servos, a phone for eyes, and 100–200 demonstrations. A ball and a block sit on the plate; you type "pick up the ball" or "pick up the block", and the arm drops that one in the cup. The words decide which object it goes for.

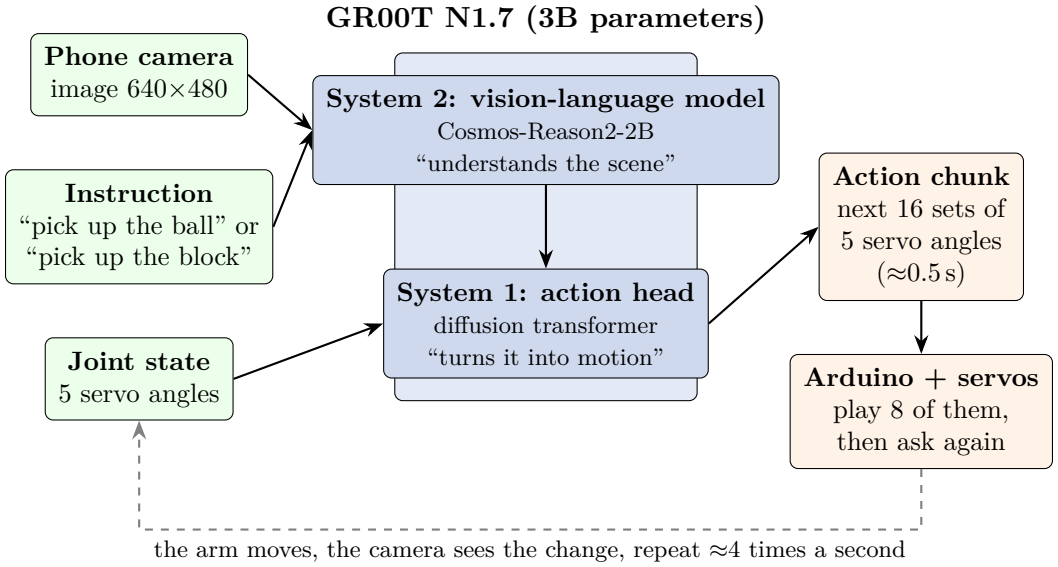


Figure 1: The control loop. Everything on the left is produced by this arm; the middle is GR00T; the right is sent back to the arm.

2 What GR00T is, in plain words

A foundation model for robots. Language models are pretrained on the internet and then adapted to a task. GR00T does the same for robot *motion*: it is pretrained on a mix of real robot recordings, simulated robots and videos of humans, then *post-trained* (fine-tuned) on a small set of demonstrations of a specific robot doing a specific task [1, 2].

Two systems, like fast and slow thinking. The **System 2** part is a vision-language model: it looks at the camera image and reads the instruction, and produces a rich description of the situation. The **System 1** part is an *action head*, a diffusion transformer: it takes that description plus the robot's current joint angles and produces motor commands. System 2 is the "what is going on and what should happen", System 1 is the "how do my joints get there".

It outputs a short plan, not one move. The action head predicts an **action chunk**: the next 16 commands (about half a second). Predicting a chunk makes motions smooth and lets the model commit to a plan instead of dithering every frame. We play the first 8 and then ask again with a fresh image, so the arm keeps correcting itself.

How it generates actions: denoising. The action head starts from random numbers and refines them over a few steps (4 by default) into a sensible chunk, guided by what System 2 saw. This "flow matching" / diffusion trick is the same family of method image generators use, applied to joint angles instead of pixels.

One brain, many bodies. The big shared network is the same for every robot. What is specific to a body is small: how its joint state is read in and how actions are read out. GR00T has a reserved slot for robots it has never seen, `NEW_EMBODIMENT`. Our arm is registered there with 4 arm joints, 1 gripper and 1 camera.

3 Every GR00T concept, and where it lives on this arm

Table 1: Mapping GR00T's pieces onto the arm. File names are in the GitHub repo.

GR00T concept	On this arm	Where
Vision input	One phone camera on a fixed mast looking down at the work area (key `front`). Fixed so every demo looks the same.	mast + cradle STLs, `arm_io.py`
Language instruction	"pick up the ball" or "pick up the block", saved with each demo; typed in at run time	`recorder.py`, `run_policy.py`
Proprioceptive state	5 servo angles in degrees: base, shoulder, elbow, wrist (`single_arm`) and gripper (`gripper`)	`export_dataset.py`
Embodiment	`NEW_EMBODIMENT` with our modality config	`gr00t_arm_config.py`
Dataset format	LeRobot v2.1 + `meta/modality.json` (parquet tables + mp4 videos)	`export_dataset.py`
Action chunk	16 future commands; we execute 8 then re-query	`run_policy.py`
Relative vs absolute actions	Arm joints are predicted as *changes* from the current angles (generalises better); the gripper as an absolute open/close target	`gr00t_arm_config.py`
Post-training	Fine-tune on 100–200 demos on a rented cloud GPU, ≈3000 steps	`gr00t/finetune.sh`
Inference	Model runs on the cloud GPU as a server; the laptop sends image + state, gets actions back over ZeroMQ	`gr00t/serve.sh`, `policy_client.py`
Demonstrations	Scripted pick-and-place (auto-collected; random layout, ball or block about half each) or keyboard teleop	`autodemo.py`, `teleop.py`

4 The whole pipeline

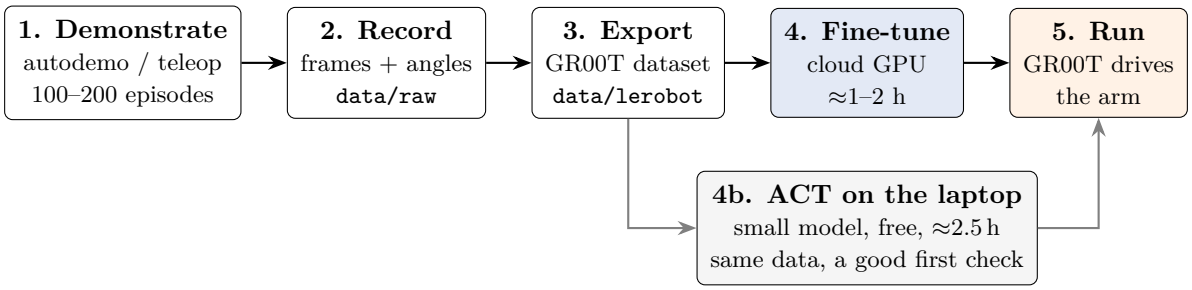


Figure 2: From demonstrations to a running policy. ACT [3] is a much smaller imitation-learning model that trains on the laptop; it checks the data before paying for a GPU.

5 Presenting it in a humanoids talk

5.1 A 3-minute script

What to say

1. **The problem.** "A humanoid has to turn what it sees and what it's told into motor commands for dozens of joints. Writing that by hand doesn't scale. NVIDIA's answer is GR00T, a foundation model for robots."
2. **Point at the arm.** "This is a $10 open-source arm, mobiobi's Easy Robot Arm: five hobby servos and my phone as the camera. GR00T doesn't care that it's small. The recipe is the same one used for humanoids."
3. **Inputs and outputs** (Fig. 1). "Three inputs: the camera image, an instruction in English, and the current joint angles. One output: a chunk of the next 16 joint commands, about half a second of motion."
4. **Two systems.** "Inside, a vision-language model figures out *what* to do, and a diffusion transformer turns that into *how* the joints move. Slow thinking and fast moving."
5. **New body.** "GR00T has never seen this arm. I registered it as a new embodiment: 4 joints, a gripper, one camera, and fine-tuned it on about 150 demonstrations, half picking the ball and half the block, each with its sentence. That took an hour or two on a rented GPU."
6. **Demo** (Section 5.2).
7. **Back to humanoids.** "A humanoid is the same loop with more joints, more cameras and much more data. Pretraining on many bodies is what lets each new body learn from only a handful of demonstrations."

5.2 Live demo plan

Pick the most ambitious option you have working, and keep the next one ready.

1. **GR00T live.** Start `serve.sh` on the cloud GPU before class, connect with `run_policy.py gr00t`. Needs internet. Put the ball and the block on two marked spots, ask the audience which one, and type it: `pick up the block`. When it's done it asks again, so swap the objects around and do the other one.
2. **ACT live (no internet).** Same pipeline, smaller model running on the laptop: `run_policy.py act`. Say clearly that this is the small model trained on the same data. ACT can't read words: it gets "ball" or "block" as a single number (0 or 1). That difference is a good point to make: GR00T understands the sentence, ACT only gets a switch.
3. **Scripted demo.** `autodemo.py -target block -spot C1 -other-spot A3` always works. Explain that this script is how the training demonstrations were collected, and that the learned policy never sees the spot names, only the camera image and the words.
4. **Video.** Screen-record a successful GR00T run the day before as a backup.

Show the dataset too: open one recorded `episode_000000.mp4` to make "demonstration data" concrete.

5.3 Questions you might get

Why not just program the pick-up? You can, and the scripted demo does. But that only works if the ball is exactly on a known spot. A learned policy reads the image, so the same approach scales to tasks

nobody can script, like folding a shirt.

How much data? 100–200 demonstrations here, about half for each object. Pretraining is what makes that enough.

Does it really understand the words? Both objects are on the plate in every demo, so the only way to know which one to pick is the instruction. If it picks the right one after you swap their places, it's using the words and the image together.

Why chunks of actions? Smoother motion, fewer model calls, and the model commits to a plan. Replanning every 8 steps still lets it correct mistakes.

Does it generalise? Within reason: other spots on the plate, small lighting changes. Not to a new task it was never shown.

Why the cloud? Running GR00T needs a GPU with about 16 GB of memory and fine-tuning about 40 GB; the laptop has 8 GB.

Part II
Build and run instructions

6 What you need

Table 2: Parts. Nothing exotic; most of it you already have. Beyond these, the only possible cost is cloud GPU time for GR00T, which has free options (Section 15).

Part	Qty	Notes
SG90 micro servo (with horns and screws)	5	base, shoulder, elbow, wrist, gripper. The arm's design has room for a 6th (wrist roll); we leave it out
Arduino Nano + USB cable	1	the cable must carry data, not only charge
Adjustable bench power supply	1	set to 5.5 V, current limit 3 A
Breadboard + male–male jumper wires	–	to share power between servos
Superglue (CA)	–	optional: only if a part slips on its horn or the gripper turns
Orange ping-pong ball (40 mm)	1	or print two `12_ball_half` and glue them
Printed block (25 mm cube)	1	`13_block`, in a colour that stands out from the ball and the plate (blue, with an orange ball on a light grey plate)
Phone + webcam app	1	Phone Link (Android) or Iriun / DroidCam (any phone)
PLA filament	a few hundred g	Bambu printer; one 1 kg spool is plenty
Optional: 470–1000 μF capacitor	1	across the servo power rail; smooths current spikes
Optional: small elastic bands	–	the author's tip: around the joints to damp vibration

7 Printing

The arm is mobiobi's *Easy Robot Arm* [4], printed without changes from the author's files in `cad/stl_opensource/easy_robot_arm/`. The work cell (baseplate, camera mast, phone cradle, cup) is ours, in `cad/stl`, generated from `cad/arm.py`.

Table 3: Print list. Arm parts: the author's settings, PLA, 0.2 mm layers, supports on, 10% gyroid infill. Work-cell parts: PLA, 0.2 mm, 3 walls, 20% infill, no supports.

File	Qty	What it is / how to print
The arm (`cad/stl_opensource/easy_robot_arm`)		
`Base`	1	drum; the base servo sits inside
`BaseRotator`	1	turntable on the base servo; holds the shoulder servo
`BaseCap`	1	screw-on ring that holds the turntable on the drum
`Arm60mm`	2	upper arm and forearm (each carries the next servo)
`offAxis_gTrunkUndr`	1	wrist piece; hangs the gripper below the wrist axis
`gripperMount`, `gripperCap`	1 each	hold the gripper servo; the cap clamps it to the wrist piece
`gripperOne`, `gripper_Two`	1 each	geared fingers (they open sideways)
`gripperPinOne`, `gripperPinTwo`	1 each	finger pivot pins
The work cell (`cad/stl`)		
`07_baseplate`	1	flat; 250×200 mm, the longest print (15% infill is fine)
`08_camera_mast`	1	standing up, 4 walls
`09_phone_cradle`	1	back face down (as exported)
`10_pin`	1	lying flat
`11_cup`	1	upright
`12_ball_half`	2	only if you have no ping-pong ball; open side down
`13_block`	1	the second object; contrasting colour, 15% infill

Not needed: `BaseLeg` (the drum stands in a ring on the baseplate instead), `Arm75mm`, `Arm90mm`, `gripperTrunk`, `offAxis_gTrunk`, `twistMount` and the two `28BYJ-48` stepper parts. The `00`–`06` files in `cad/stl` are an older arm design and aren't used.

8 Laptop software setup

1. Install **uv** (Python manager) and **Arduino CLI** in PowerShell:

```
powershell -c "irm https://astral.sh/uv/install.ps1 | iex"
winget install ArduinoSA.CLI
```

2. Get the code and install the Python packages:

```
gh repo clone Yishane/gr00t-arm
cd gr00t-arm\software
uv sync
```

3. Phone as webcam: install **Iriun Webcam** or **DroidCam** on the phone and the PC (or on Android use Windows Phone Link → "Use as connected camera"). It shows up as a normal webcam. The laptop's own camera is usually index 0 and the phone 1 or 2.

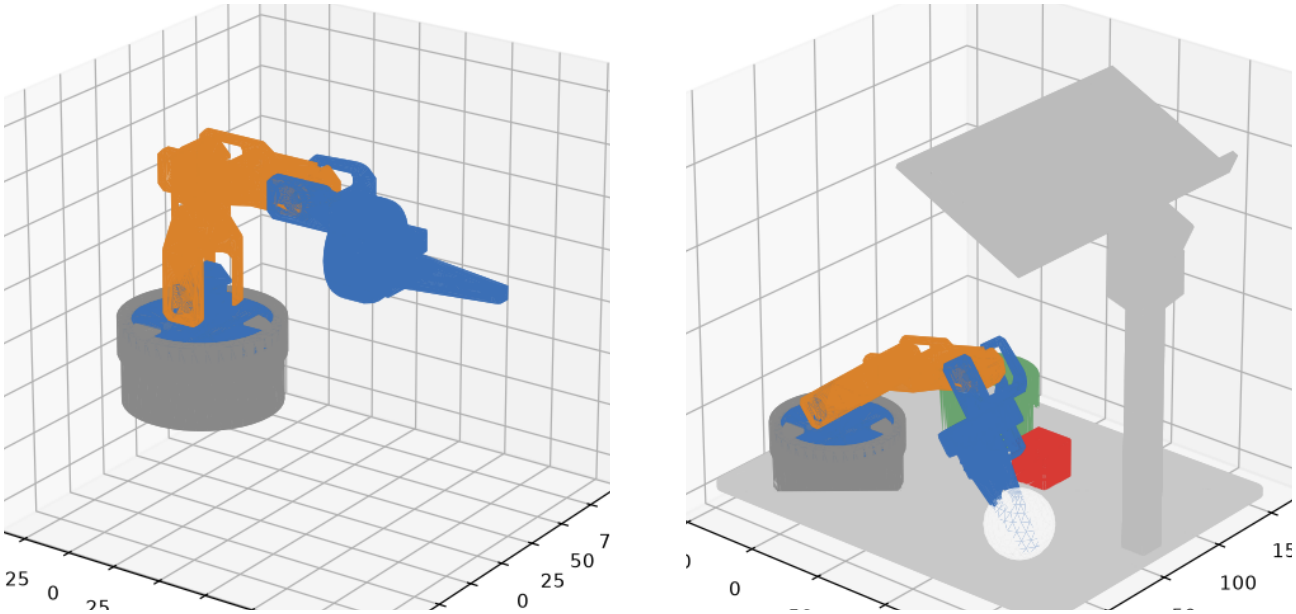


Figure 3: Left: the Easy Robot Arm in its home pose, built from the author's 3D model with two 60 mm links. Right: the work cell with the baseplate, cup and phone camera mast, the arm gripping the ball on C1 with the block on A3. `cad/assembly.py` checks the arm against itself and the work cell: every marked spot (ball and block), the lift above it and the drop over the cup are clear, and the gripper never touches the other object in any layout the demos use.

9 Wiring

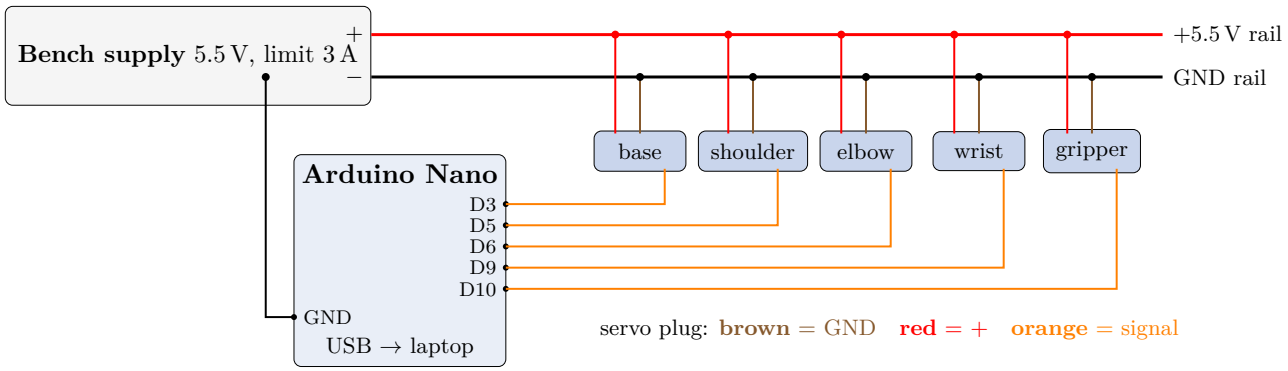


Figure 4: Wiring. Servo power comes from the bench supply, *never* from the Nano's 5V pin. The Nano's GND must connect to the supply's negative so the signals have a common reference. Orange signal wires go to D3, D5, D6, D9, D10 in the order base, shoulder, elbow, wrist, gripper.

1. Set the bench supply to **5.5 V** with the current limit at **3 A**, output off.
2. Bench supply + to the breadboard's red rail, − to the blue rail. (Optional: the capacitor across the two rails, its stripe/negative leg to the blue rail.)
3. Each servo: red to the red rail, brown to the blue rail (use jumpers into the servo's 3-pin plug), orange to its Nano pin from Fig. 4.
4. One jumper from a Nano GND pin to the blue rail.

5. Nano to the laptop by USB.

Why the separate supply

Five SG90s can pull over 2 A when they start moving together. USB gives 0.5 A: the Nano would reset and the arm would twitch. If the supply hits its current limit (the CC light comes on) something is jammed; turn it off and check.

9.1 Upload the firmware

Find the Nano's port in Device Manager → Ports (e.g. COM5), then:

```
cd gr00t-arm\firmware
.\upload.ps1 COM5
```

The script tries the "old bootloader" (most Nano clones) and then the new one. The firmware listens on USB serial for lines like `90,90,90,90,125` (five angles) and moves each servo smoothly toward its target.

10 Assembly

The author gives no written steps. The parts are made to need only what comes with the SG90s (the horns and screws), and every servo slides into a pocket so it can be swapped. Look at the photos on the Printables page [4] for how each piece sits.

The one rule

Every servo must be at **90 °** when the part it drives goes on, with that part in the **home pose**: arm facing forward, upper arm straight up, forearm level and pointing forward, gripper level and in line with the forearm, fingers closed. The software assumes this; get it right and calibration is only small trims.

1. **Center all servos.** Wire all five servos with nothing attached, power on, and run

   ```
   cd gr00t-arm\software
   uv run calibrate.py
   > center
   ```

   All five turn to 90 °. Leave them powered while fitting parts, and don't turn a shaft by hand.

2. **Base.** Base servo into the drum (`Base`), shaft up. Put the turntable (`BaseRotator`) on its horn so the arm will swing forward and back. Screw `BaseCap` over it onto the drum. Leave the `BaseLeg` feet off.

3. **Shoulder.** Shoulder servo into the turntable. The first `Arm60mm` (upper arm) onto its horn, pointing *straight up*.

4. **Elbow.** Elbow servo into the top of the upper arm. The second `Arm60mm` (forearm) onto its horn, pointing *forward, level*.

5. **Wrist.** Wrist servo into the end of the forearm. The wrist piece (`offAxis_gTrunkUndr`) onto its horn so the gripper will hang *below* the forearm, pointing forward and level.

6. **Gripper.** Gripper servo into `gripperMount`; the fingers on their pins with the gears meshed and the fingers *closed*; then clamp `gripperMount` to the wrist piece with `gripperCap`.
7. **No wrist-roll servo.** The design has room for one between the wrist piece and the gripper; we leave it out so the software stays at 5 joints. If the gripper turns on the wrist piece, put a drop of superglue between the two discs.
8. **Work cell.** Stand the drum in the ring on the baseplate (spots A1–C3 in front, the cup ring to the left) and fix it with double-sided tape or a few dots of hot glue, so the base servo turns the arm and not the drum. Push the mast into its socket, the pin into the mast top, and the phone cradle onto the pin. Clamp or tape the baseplate's back edge to the desk.
9. **Phone.** Landscape on the cradle, resting on the lip, *rear camera near the lip*. Open the webcam app.
10. **Cables.** Leave slack at every joint so nothing pulls when the arm turns.

If the shoulder struggles

The author notes the shoulder servo works hardest. If it sags or buzzes, swap just that one for an MG90S (same size, metal gears, no code changes). Small elastic bands around the joints, as in the author's photos, damp vibration.

11 Calibration

Still in `calibrate.py`:

1. `home` — the arm should go to the home pose.
2. `test` — moves each joint a little and tells you which way it should go. If one goes the wrong way, type `flip <joint>` (0 base, 1 shoulder, 2 elbow, 3 wrist).
3. If a joint is a few degrees off in the home pose: `trim <joint> <degrees>`, e.g. `trim 1 -4`.
4. Gripper: `s 4 <deg>` to try angles. Find an angle that is well open, and one that squeezes the *block* firmly (it's the smaller object, so this also grips the ball); set them with `grip open 125` and `grip closed 85` (your numbers). The fingers open sideways; check with the block and then the ball held between them. If the servo buzzes hard on the ball, tell Claude: the ball can get its own closed angle.
5. `save`, then `quit`. This writes `software/calibration.json`.

Check the tool point

The arm's lengths in `software/config.py` were measured from the author's 3D model, not from your printed arm. In Section 12, if the fingers close a little in front of, behind, above or below the ball, tell Claude how far off it is; `L_TOOL` (along the gripper) and `TOOL_UP` (how far the fingers hang below the wrist) get a small trim.

12 First moves

```
uv run autodemo.py --spot B2 --camera 1
```

Put the ball on spot B2 first. The arm should pick it up and drop it in the cup. Then the block, with the ball on another spot:

```
uv run autodemo.py --target block --spot C1 --other-spot A3 --camera 1
```

Turn the block so one face points at the arm's base: the fingers close sideways. Then try driving it yourself:

```
uv run teleop.py --camera 1
```

Click the window, then: W/S forward/back, A/D left/right, R/F up/down, Q/E tilt the gripper, Space open/close, H home, Esc quit. If it says "can't reach that", the target is outside the arm's reach or would hit the plate; it refuses rather than crashing. Add `-sim` to any script to try it without hardware.

13 Collecting demonstrations

1. **Automatic (recommended):**

   ```
   uv run autodemo.py --record 150 --camera 1
   ```

   Each time it tells you where to put the ball and the block and which one it will pick, e.g. `block on C1, ball on A3 -> "pick up the block"`. Place both, press Enter, and it picks and places while recording, saving the instruction with the demo. It picks the ball or the block about half the time each. Aim for 100–200 demos (about 20 seconds each, so 150 take under an hour; split it into sessions, it keeps numbering where it left off).

2. **By hand (optional, adds variety):** in `teleop.py`, T switches the instruction between "pick up the ball" and "pick up the block" (shown at the top of the window), Enter starts a recording, Enter again saves it, Backspace throws it away. Always pick the object the instruction names.

Good data

Keep the camera, lighting and cup exactly where they will be during the demo. Both objects are on the plate in every demo; the middle spot B2 isn't used because it's too close to the others. Place the objects a little off-center sometimes, with a block face towards the base. Delete any failed demo folder from `data/raw/pick_ball`.

Then export for GR00T:

```
uv run export_dataset.py
```

This writes `data/lerobot/pick_ball` (Table 1). GR00T's own data loader was used to check this format.

14 Training ACT on the laptop (free, optional but recommended)

```
cd software\act
uv sync
```

```
uv run export_act.py
uv run train_act.py --steps 20000
uv run ..\run_policy.py act --camera 1 ^
    --checkpoint ..\..\outputs\act\pick_ball\checkpoints\020000\pretrained_model
```

Training runs at about 2.3 steps per second on the RTX 5050 (about 2.5 hours for 20k steps). It asks for an instruction; type `pick up the ball` or `pick up the block` (or just `ball` / `block`). Each runs for 20 seconds (`-seconds`), then the arm goes home and it asks again; `q` quits. If ACT picks up the right object, the data is good and GR00T will learn from it too.

15 Fine-tuning and running GR00T in the cloud

15.1 One-time accounts (you do these)

1. **Hugging Face**: make a free account at https://huggingface.co, open https://huggingface.co/nvidia/Cosmos-Reason2-2B and request access (GR00T is built on it). Then create an access token (Settings → Access Tokens, "read").
2. **GPU time.** Fine-tuning needs a GPU with at least 40 GB (L40S, A100 or H100). Try free credits first. **Lightning AI**'s free plan gives monthly credits and a Studio you can SSH into, so the steps below work unchanged. **Modal**'s free plan gives about $30 a month, but it needs a small wrapper script first. Otherwise, rent a machine on a paid cloud such as RunPod or Lambda. Free 16 GB GPUs (Colab, Kaggle) are too small.

This is the only step that can cost money

Everything before this section is free. If you don't want to pay and there are no free credits, stop after ACT: it uses the same data and the same control loop, and it is a fine live demo (Section 5.2, option 2). Check the free plans' current terms on their pricing pages before signing up.

15.2 Steps

1. Start a Linux GPU machine (Ubuntu, L40S/A100/H100, ≥100 GB disk). Note its SSH address and port.
2. Copy the GR00T scripts and the dataset up (from the `gr00t-arm` folder):

```
scp -P <port> -r software\gr00t data\lerobot\pick_ball root@<ip>:~/
```

3. Log in and set up (installs GR00T and downloads the base model, ≈20 min):

```
ssh -p <port> root@<ip>
export HF_TOKEN=hf_your_token
bash ~/gr00t/setup_cloud.sh
```

4. Fine-tune (about 1–2 hours; use `tmux` so it survives a disconnect):

```
tmux
bash ~/gr00t/finetune.sh ~/pick_ball
```

At the end it plots predicted vs. recorded actions for episode 0 in `/tmp/open_loop_eval`. The lines should overlap. Flat predicted lines mean something is wrong with the data.

5. Serve the model:

```
bash ~/gr00t/serve.sh
```

6. On the laptop, open a tunnel in one terminal and run the arm in another:

```
ssh -p <port> -L 5555:localhost:5555 root@<ip>
uv run run_policy.py gr00t --host localhost --camera 1
```

Type the instruction at the `instruction>` prompt, the same as for ACT.

7. **Stop the cloud machine when you're done**: it bills by the hour.

Cost and time

A paid cloud GPU costs roughly $1–3 per hour. Setup plus one fine-tune plus testing is a few hours, so budget $10–20 for the first run, or use free credits if they cover it. Download the checkpoint folder (`~/gr00t_arm_ckpt`) if you want to reuse it later without retraining.

16 Troubleshooting

Symptom	Fix
"No Arduino found"	Check the USB cable carries data; pass `-port COM5`.
Upload fails	Try the other bootloader (the script does both); close other programs using the port.
Servos twitch / Nano resets	Servo power must come from the supply; Nano GND tied to supply −; add the capacitor.
A joint moves the wrong way	`calibrate.py`: `test`, then `flip <joint>`.
Arm sags or buzzes holding pose	Normal for SG90s near their limit, mostly at the shoulder; keep the gripper light. An MG90S at the shoulder is a drop-in upgrade.
The drum turns instead of the arm	Fix the drum in its baseplate ring with tape or hot glue.
"Camera N didn't open"	Try `-camera 0`, `1`, `2`; make sure the phone app is running.
"can't reach that" in teleop	The target is out of reach or the gripper would hit the plate. Move back.
GR00T: `GatedRepoError`	Your HF account hasn't been granted Cosmos-Reason2-2B access yet, or `HF_TOKEN` is wrong.
`run_policy`: server not reachable	Is `serve.sh` running and the SSH tunnel open?
Policy runs but misses	More demos, more varied object positions, same camera position as in training.
Picks the wrong object	More demos (both objects about equally), and check the instructions saved in `meta.json` match what was picked.

17 Where everything is

`cad/` `arm.py` (work-cell parts), `assembly.py` (collision and reach checks on the author's model), `stl/`, `renders/`, and the arm itself (mobiobi's files, CC BY-NC-SA, with `ATTRIBUTION.md`) in `stl_opensource/easy_robot_arm/`

firmware/ arm/arm.ino, upload.ps1

software/ config.py (geometry, calibration), kinematics.py (+ tests), calibrate.py, teleop.py, autodemo.py, recorder.py, export_dataset.py, policy_client.py, run_policy.py

software/gr00t/ gr00t_arm_config.py, setup_cloud.sh, finetune.sh, serve.sh

software/act/ export_act.py, train_act.py

context.md running notes on the project's state

References

[1] NVIDIA et al., "GR00T N1: An Open Foundation Model for Generalist Humanoid Robots," arXiv:2503.14734, 2025.

[2] NVIDIA, "Isaac GR00T" (code, GR00T N1.7 model, fine-tuning guides), https://github.com/NVIDIA/Isaac-GR00T.

[3] T. Z. Zhao, V. Kumar, S. Levine, C. Finn, "Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware," RSS 2023, arXiv:2304.13705.

[4] mobiobi, "Easy Robot Arm (SG90 servos & now steppers too)," Printables model 97370, CC BY-NC-SA, https://www.printables.com/model/97370-easy-robot-arm-sg90-servos-now-steppers-too-update.

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